Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 12

Subject: Physics

Medium: English

Academic Year 2026-27

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This book is prepared as a teaching guide for teachers.

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Unit 1: Transformers: Step-Up, Step-Down and Power Transmission

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Unit 1: Transformers: Step-Up, Step-Down and Power Transmission

Learning Objectives

• Students will explain how a transformer transfers energy between two coils.

• Students will use the turns ratio to calculate output voltage.

• Students will explain why high voltage is used for long-distance power transmission.

Key Concepts

• A transformer transfers electrical energy between two coils through a changing magnetic field, with no moving parts.

• The turns ratio sets the voltage ratio: output voltage equals input voltage times secondary turns divided by primary turns.

• A step-up transformer raises voltage and lowers current; a step-down transformer lowers voltage and raises current.

• Transformers need alternating current because only a changing current produces the changing magnetic flux that induces voltage.

• Power stations step voltage up for long-distance transmission so that less energy is lost as heat in the wires, and it is stepped down again before reaching homes.

Teaching Methodology (Step by Step)

1. Ask students why a phone charger is warm and much smaller than the wall socket voltage it plugs into.

2. Draw two coils on a shared iron core and show how a changing current in one induces a voltage in the other.

3. Work the turns-ratio formula with numbers on the board for a step-up and a step-down case.

4. Explain why direct current cannot work a transformer: no change, no induced voltage.

5. Tell the journey of electricity from power station to home, naming each step-up and step-down.

6. Students solve two turns-ratio problems and check each other's answers.

Classroom Activities

• Turns-ratio race: groups compute the output voltage for different turns ratios.

• Transformer hunt: list three places where a transformer is used in daily life.

• Power-line story: trace electricity from a power station to a home plug, naming each voltage change.

Assessment Questions

1. A transformer has 200 primary turns and 1000 secondary turns. If the input is 220 V, what is the output voltage?

2. Why can transformers work only with alternating current, not direct current?

3. Why is electricity sent over long distances at very high voltage?

Homework

• Explain with a diagram why a mobile-phone charger contains a step-down transformer.

• A step-up transformer has 100 primary turns and 2000 secondary turns with a 240 V input.

Calculate the output voltage.

Time Allocation

40 minutes

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